

DWARAKANATH C

GCP Cloud Developer & AI Engineer | Senior Backend Engineer (Golang / Java)
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PROFESSIONAL SUMMARY

Senior Backend Software Engineer and GCP Cloud Developer with 10+ years of experience across backend development, event-driven microservices, and cloud/AI engineering, including 4+ years of hands-on Golang and 6+ years in Java/J2EE. Builds high-throughput distributed systems with Go, Kafka, RabbitMQ, PostgreSQL, Redis, and Docker/Kubernetes for regulated platforms in financial services, travel, and FMCG. Currently designs production AI platforms on GCP using Vertex AI, Gemini, Agent Engine, Document AI, RAG, BigQuery, Cloud Run, Pub/Sub, and Terraform across document intelligence, sales prioritisation, and cash allocation. Comfortable working across time zones and contributing to architecture, code quality, observability, and delivery.

TECHNICAL SKILLS

- Languages: Golang, Java (Spring Boot, JPA), Python, SQL
- Backend & APIs: RESTful services (net/http, Gorilla Mux, Gin, Echo, Spring Boot)
- Event-Driven Systems: RabbitMQ, Kafka, Pub/Sub (message queues, event processing, async workflows)
- Databases & Cache: PostgreSQL, Redis, MongoDB, MySQL, Oracle, BigQuery
- Containers & Orchestration: Docker, Kubernetes (deployment, health checks, resource limits, autoscaling)
- Cloud & DevOps: Google Cloud Platform (Cloud Run, BigQuery, Pub/Sub, IAM, Terraform, API Gateway, Cloud Scheduler, Secret Manager), AWS, OpenShift Hydra, Jenkins, Spinnaker
- AI & Data: Agentic AI, RAG, Vertex AI, Agent Engine, Gemini, Google ADK, Document AI, machine learning, LLM inference (vLLM, Gemma), BigQuery SQL, QlikCloud
- Quality & Observability: Unit testing, BDD (Cucumber), Zap Logger, Splunk, Elastic, Kibana
- Practices: Microservices, Event-Driven Architecture, Clean Architecture, Agile/Scrum, Code Reviews

EDUCATION

Master's in Computer Application (M.C.A) - SVU, India - 2012

Bachelor's in Computer Science (BSc) - SVU, India - 2009

PROFESSIONAL EXPERIENCE

Project 1: Cloud & AI Platform Engineering - Rentokil Initial, UK | Team size: 8 | 03/2026 to current

Role: GCP Cloud Developer & AI Engineer

- **Consumables - Document Intelligence:** Designed an event-driven platform that converts incoming safety and chemical PDFs into structured analytics using Drive webhooks, API Gateway, Cloud Run, Pub/Sub, Document AI, BigQuery, Cloud Storage, and QlikCloud.
- **Consumables - Reliability and RAG:** Separated fast webhook ingestion from long-running AI workers, added RAG over extracted BigQuery entities and layered dead-letter recovery across Cloud Storage, BigQuery error rows, and Pub/Sub retry topics.
- **EU Prospect Prioritisation:** Delivered a three-CTE BigQuery model covering employee, occupation, and prospect data so ranking includes only leads assigned through employee lead codes.
- **Multi-agent Sales Intelligence:** Built Vertex AI Agent Engine services for data retrieval and RAG enrichment, ML-based value estimation, and prospect prioritisation, orchestrated with Gemini and Google ADK.
- **AI Cash Allocation:** Built an event-driven credit-control workflow that classifies inbound Gmail messages with Gemini on Vertex AI and writes structured results to a BigQuery audit trail.
- **AI Cash Allocation - Agentic Matching:** Designed a RAG workflow over invoice history to match remittances, propose cash allocations, and route processing through Pub/Sub and Cloud Run.
- **LLM Inference PoC:** Led a self-hosted Gemma deployment on Cloud Run GPU with vLLM to evaluate a cost-efficient alternative to third-party inference APIs.

- **Platform Quality:** Improved reliability through Terraform-managed infrastructure, autoscaling, data-quality reviews, alerting, code reviews, and documented recovery paths.

Tech: GCP (Vertex AI, Agent Engine, Gemini, Document AI, BigQuery, Cloud Run, Pub/Sub, Cloud Storage, IAM, API Gateway, Cloud Scheduler, Secret Manager), RAG, Google ADK, Terraform, Python, SQL, QlikCloud, vLLM, Gemma

Project 2: Global Loyalty Program - Amex, UK | Team size: 7 | 08/2024 to 04/2026

Role: Senior Golang Developer

- Design and develop scalable Golang microservices supporting loyalty enrolments and reward processing platforms, handling millions of transactions daily.
- Build RESTful APIs using net/http and Gorilla Mux, focusing on clean design and maintainability with comprehensive unit test coverage.
- Implement event-driven workflows using Kafka to process lifecycle, eligibility, and reward events with guaranteed delivery mechanisms.
- Optimise services for high throughput (10,000+ requests/minute), low latency, and resilience in distributed systems.
- Work with PostgreSQL, MongoDB, and Redis for transactional data handling, caching strategies, and performance optimization.
- Deploy and manage containerised services using Docker and Kubernetes, including configuration of health checks, resource limits, and auto-scaling policies.
- Lead architecture discussions for microservice decomposition, API design patterns, and event-driven messaging strategies across cross-functional teams.
- Write comprehensive unit tests and BDD test suites, perform code reviews, and support production incident resolution with root cause analysis.

Tech: Golang, Kafka, PostgreSQL, Redis, Docker, Kubernetes, Microservices, Git, Elastic, Kibana, OpenShift Hydra

Project 3: Campaign Scheduler - Expedia, US | Team size: 6 | 05/2022 to 07/2024

Role: Golang Developer

- Developed backend services in Golang for campaign scheduling and customer engagement platforms processing 500,000+ scheduled events daily.
- Implemented asynchronous messaging using RabbitMQ for job queues, event processing, and inter-service communication, handling 100,000+ messages/day with guaranteed delivery and retry mechanisms.
- Designed and integrated event-driven workflows for communication and calendar scheduling services using RabbitMQ exchanges and routing patterns.
- Implemented concurrent and asynchronous processing using goroutines and channels to handle high-volume scheduling workloads efficiently.
- Designed and developed RESTful APIs using Echo and net/http with comprehensive error handling and input validation.
- Built timezone-aware scheduling logic for global users supporting 50+ timezones with daylight saving adjustments.
- Deployed containerised services using Docker on AWS infrastructure with CI/CD pipelines via Jenkins.
- Performed code reviews and contributed to improving service stability and performance through observability implementations.

Tech: Golang, RabbitMQ, Microservices, Echo, net/http, AWS, Docker, MongoDB, Jenkins

Project 4: Relocation Project - Aires Mobility, US | Team size: 6 | 09/2021 to 05/2022

Role: Golang Developer

- Developed backend APIs for a Benefits Management System using Golang with focus on clean architecture principles.
- Designed microservices architecture, API contracts, and data models from scratch following RESTful best practices.
- Authored comprehensive technical documentation and implemented unit and integration tests achieving 85%+ code coverage.
- Participated in sprint planning, daily standups, and peer code reviews in an Agile environment.

Tech: *Golang, Microservices, AWS, Docker, PostgreSQL*

Project 5: iGTB Payment Service Hub (IPSH) - Santander, Spain | *Team size: 10 | 03/2019 to 08/2021*

Role: Software Developer

- Developed RESTful backend services using Java and Spring Boot for enterprise banking and payment systems in a highly regulated financial environment.
- Implemented business validation rules using Drools Engine ensuring compliance with financial regulations and industry standards.
- Worked on country-specific data mappings and regulatory compliance features including SEPA, SWIFT, and local payment schemes.
- Followed Agile and TDD practices, collaborating closely with QA, compliance, and business teams to ensure quality delivery.

Tech: *Java 8, Spring Boot, JPA, Microservices, Docker, AWS, Swagger, Git, PostgreSQL*

PERSONAL PROJECTS

- **LifeHub (In progress):** Built an end-to-end personal productivity platform with FastAPI, PostgreSQL, Gemini priority classification, Next.js, and Expo; the Todos workflow is complete, with Calendar, meeting notes, and Google OAuth planned next.
- **Maths App (Shipped v1):** Developed a Grade 1 Android learning app in Kotlin and Jetpack Compose covering place value, expanded form, ordinals, addition, and subtraction, with Learn/Practice modes and instant scoring feedback.
- **The AI Pulse (Shipping daily):** Built a daily AI-engineering brief pipeline in Python using Groq, OpenRouter, LLM relevance scoring, and Gmail delivery to produce concise builder- and leadership-focused summaries.

Tech: *Python, FastAPI, PostgreSQL, Gemini, Next.js, TypeScript, Expo, Kotlin, Jetpack Compose, Groq, OpenRouter, Gmail*

AWARDS AND CERTIFICATIONS

- Best Performer of the Month Award from SmartLoyal
- AWS Certified Developer Associate - Dec 2023
- GCP Professional Cloud Architect (in progress)
- Spring Boot and Microservices with Spring Cloud (Udemy)
- GEM Award from Polaris for Excellent Performance
- Spot Excellence Award from Polaris
- Agile Software Development (Coursera)

EARLIER PROJECTS

- Interac-e-Transfer (Mobile Application) as Developer: 07/2017 – 02/2019
- Branch Profitable Index (BPI) as Developer: 09/2015 to 06/2017